Oliver Soeser

Edinburgh, United Kingdom hello@oliversoeser.com www.oliversoeser.com oliversoeser

About

I am a final-year undergraduate studying Computer Science and Mathematics and am highly motivated to pursue research in programming languages and theoretical computer science. My experience stems from the internships I undertook at ETH Zürich and IST Austria, where I got involved with programming language research by contributing to the Iris separation logic framework in the Lean theorem prover. I am passionate for the use of rigorous mathematics to understanding the fundamentals of computer programming, in particular in its connections with algebra and category theory. My research interests include **(dependent) type theory**, **interactive theorem proving**, **automated reasoning**, **formal logic**, **program verification**, **denotational semantics**, **domain theory**, **universal algebra**, and **applied category theory**.

Experience

Summer Research Fellow *July 2026 – August 2026*
Programming Language Foundations Lab, ETH Zürich
Supervisors: Prof. Ralf Jung, Max Vistrup

- Implemented Lean language keywords for defining recursive and inductive predicates within the Iris separation logic using fixpoints, making definitions within the logic as simple as in the meta-logic.
- Built fully automated proof procedures for fixpoint conditions using typeclasses and heuristics.
- Studied category-theoretic generalisations of domain theory used for the Iris model.
- Highly competitive fellowship with an acceptance rate under 1%.

Scientific Intern *June 2025 – August 2025*
Programming Languages and Verification Group, IST Austria
Supervisor: Prof. Michael Sammler

- Contributed to porting the Iris project for program verification from Rocq to Lean, making use of its metaprogramming and automation capabilities.
- Implemented the crucial `iapply` and `irevert` tactics for interactive proofs in separation logic, since used over 1000 times across the project.
- Studied the foundations of programming language theory and read contemporary papers in the field.

Education

University of Edinburgh *September 2023 – May 2027*
Computer Science and Mathematics (BSc Hons)

- Topics studied include programming language theory, theoretical computer science, and abstract algebra.
- Dissertation will be supervised by *Dr Paul Jackson* and be about the **application of the Lean theorem prover to programming languages**.
- Attending talks, workshops, and seminars on current programming language research.
- Consistent top marks, on track for a first-class degree.

Teaching

Teaching Support Provider *September 2024 – Current*
School of Informatics, University of Edinburgh

- **Introduction to Computation** (INF1A): *Teaching Assistant*, *Tutor*, and *Marker*
- **Object Oriented Programming** (INF1B): *Tutor*, *Marker*, and *Lab Demonstrator*
- **Introduction to Algorithms and Data Structures** (INF2-IADS): *Tutor* and *Lab Demonstrator*

Appendix A: Academic Record

Course	**Mark (%)**	**Grade**
Compiling Techniques	80	A
Honours Complex Variables	82	A
Elements of Programming Languages	95	A
Introduction to Theoretical Computer Science	100	A
Algorithms and Data Structures	79	A
Informatics Large Practical	76	A
Combinatorics and Graph Theory	82	A
Introduction to Algorithms and Data Structures	80	A
Fundamentals of Pure Mathematics	84	A
Foundations of Data Science	81	A
Statistics	74	A
Computing and Numerics	73	A
Probability	95	A
Facets of Mathematics	84	A
Several Variable Calculus and Differential Equations	79	A
Calculus and its Applications	88	A
Proofs and Problem Solving	85	A
Object Oriented Programming	94	A
Philosophy of Science	67	B
Introduction to Linear Algebra	97	A
Introduction to Computation	100	A

Note that my grade for *Honours Algebra* is still pending as exam feedback has been delayed as part of the Marking and Assessment Boycott. My mark in the coursework component of the course was 90%.

Alongside my dissertation, in the final year of my degree I am taking *Topics in Programming Languages and Semantics*, *Commutative Algebra*, *Topics in Ring and Representation Theory*, *Introduction to Quantum Computing*, *Introduction to Number Theory*, and *Analytic Number Theory*.